Task - 3: Networking Basics for Cyber Security

Internship Task – Elevate Labs

This task focused on understanding basic networking concepts and analyzing network traffic using packet sniffing tools. The main objective was to learn how data moves across a network and how cyber security professionals monitor and analyze that traffic.

1. Basic Networking Concepts

Important concepts learned: - IP Address - MAC Address - DNS - TCP and UDP protocols

These are fundamental for understanding communication between systems.

2. Packet Capturing with Wireshark

Wireshark was installed and used to capture live network traffic. Different packets moving through the network interface were observed in real time.

3. Filtering Packets

Filters such as HTTP, DNS, and TCP were applied to analyze specific protocols. This helped in focusing on relevant network activity instead of all traffic.

4. TCP Three-Way Handshake

The process of connection establishment was observed: - SYN - SYN-ACK - ACK

This showed how reliable TCP communication is created.

5. Encrypted vs Plain Text Traffic

HTTP traffic was seen in plain text while HTTPS appeared encrypted. This proved why HTTPS is more secure for sensitive data.

6. DNS Packet Analysis

DNS queries and responses were captured. This demonstrated how domain names are converted into IP addresses before communication begins.

7. Saving Packet Captures

Packet capture files were saved for later study and analysis. This is useful in real-world incident investigation.

8. Final Observations

- Many background services communicate continuously. - Encryption protects confidential data. - Packet sniffing is essential for network monitoring.

Learning Outcome

This task improved my understanding of networking and cyber security. I learned how to analyze network traffic, identify protocols, and observe secure communication in practice.